Firestore setup guide

This guide configures the **server-side** Firestore backend. It does not perform a live migration or change production data. Keep `STORE_BACKEND=json` until the Firestore adapter has been tested against a disposable project or the local emulator.

Environment

Set these variables only in the server/deployment environment:

```
STORE_BACKEND=firestore
FIREBASE_PROJECT_ID=your-gcp-project-id
# Optional for local development:
# FIRESTORE_EMULATOR_HOST=127.0.0.1:8080
```

The Firebase Admin SDK uses Application Default Credentials (ADC). In a Google-managed runtime, use the runtime service account. For local development or another server environment, set `GOOGLE_APPLICATION_CREDENTIALS` to the absolute path of a server-only service-account JSON file:

```
GOOGLE_APPLICATION_CREDENTIALS=/absolute/path/to/service-account.json
```

Do not commit the credential file, paste its contents into `.env` or logs, or expose it through any `VITE_*` variable. `FIREBASE_PROJECT_ID` is explicit so the server cannot accidentally connect to a similarly named project.

Local emulator

From the project root, install dependencies once, then start the Firestore emulator with the disposable `leish-local` project ID (keep this terminal open):

```
pnpm install
pnpm run emulator:firestore
```

The package script pins Firebase CLI `13.35.1` for repeatable emulator startup.

In a second terminal, from the same project root, **explicitly** seed the running emulator from the existing `db_store.json` development store:

```
pnpm run seed:firestore
```

The seed command sets `FIRESTORE_EMULATOR_HOST=127.0.0.1:8080` (without `http://`) and `FIREBASE_PROJECT_ID=leish-local`. The script requires these exact values, uses no credentials, and refuses a nonempty emulator instead of overwriting data. Stop and restart the emulator without a data import to seed again. This command is never run by app startup or production deployment. The JSON file includes demo users and bcrypt password hashes; use only local development data here. To run the server against the seeded emulator, use a third terminal:

```
STORE_BACKEND=firestore FIREBASE_PROJECT_ID=leish-local FIRESTORE_EMULATOR_HOST=127.0.0.1:8080 pnpm run dev
```

The emulator is the preferred first verification target. No credentials are needed with the emulator, but the server still requires the explicit project ID.

Rules and client access

The web client does not need direct Firestore access for this application. Keep Firestore client rules deny-by-default, for example:

```
rules_version = '2';
service cloud.firestore {
  match /databases/{database}/documents {
    match /{document=**} {
      allow read, write: if false;
    }
  }
}
```

The Firebase Admin SDK used by the API is not restricted by client security rules; protect the API with its existing authentication and authorization checks and keep credentials server-only.

Switching back

Set `STORE_BACKEND=json` (or remove the variable) to use the existing local JSON store. This guide intentionally provides no migration or live-change procedure. Back up and separately plan any future data migration before changing a deployed backend.